

Complexation between Oppositely Charged Polyelectrolytes in Dilute Solution: Effects of Charge Asymmetry

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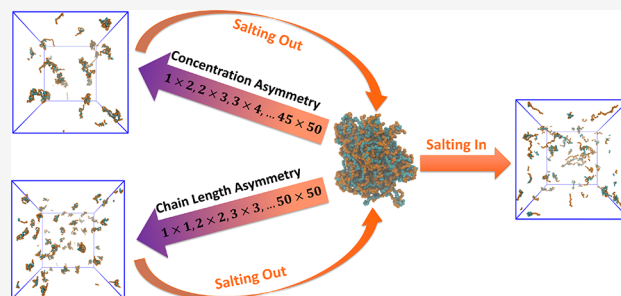


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ABSTRACT: We use dissipative particle dynamics to study polyelectrolyte complexation in dilute solutions under conditions of either chain-length or concentration asymmetry between the polycation and the polyanion, characterized by the macromolecular charge ratio of the minor component to the major component. Our results show that generally the systems form clusters carrying net macromolecular charges. The stoichiometry of these net-charged macromolecular clusters depends on the overall charge asymmetry, and clusters with certain stoichiometry can dominate the cluster population under a wide range of chain-length and concentration asymmetry. The morphology and stability of these net-charged clusters are closely related to their stoichiometry. When the charge ratio reaches a threshold value, the polyions condense into a single large coacervate cluster, signaling the onset of macroscopic coacervation. The threshold value is less for concentration asymmetry systems than for chain-length asymmetry systems at the same salt concentration. This threshold value decreases with added salt, as salt ions cause merging of small clusters into larger ones, eventually leading to the dominance of a single large cluster. This “salting-out” phenomenon is followed by a “salting-in” behavior, as further addition of salt dissolves the large cluster. The preferred size and composition of the net-charged clusters observed in the simulation highlight the significant role of length and concentration asymmetry on polyelectrolyte complex coacervation and hint at the possibility of a microstructured liquid.



INTRODUCTION

When two solutions of oppositely charged polyelectrolytes (PEs) mix under appropriate conditions, they can phase separate into a supernatant phase and a coexisting coacervate phase that contains most of the PE chains.^{1–3} This phenomenon, known as complex coacervation, has gained renewed attention in recent years for their relevance in biological membraneless organelles: they are believed to be related to the origin of life and myriads of biological functions.^{4,5} Inspired by biomimetic systems, coacervates have also been studied for novel applications such as underwater adhesion and drug/gene delivery.^{6–8} These applications and the important role of the liquidlike coacervates in biology underscore the need for a systematic understanding of the coacervation behavior.

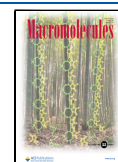
The key feature in the PE complex coacervation is the electrostatically induced liquid–liquid phase separation between the polyions and the aqueous solvent. This liquid–liquid phase separation can be macroscopic,³ microscopic,^{3,9,10} or multiphase.^{11–13} Most recent experiments, theories, and simulations focus on understanding the macroscopic PE-rich coacervate phase, as they are most relevant to applications.^{14,15} Significant progress has been made to understand the phase behavior and properties of the PE-rich phase.^{14–22} Following the pioneering Voorn–Overbeek (VO) theory²³ which highlights the role of electrostatic correlation as a driving

force for PE complex, more sophisticated theories (e.g., generalized VO,²⁴ random phase approximation (RPA),^{25–28} scaling,^{29–32} liquid-state theory,^{19,33,34} transfer matrix method,^{16,35}) along with computer simulations^{9,17,21,31,36–38} have addressed various effects, such as chain connectivity,^{17,20,33,34,39–41} solvent quality,^{42,43} excluded volume,^{17,21,26,33,34} charge sequence,^{36,37,44–52} temperature,^{53,54} and concentration asymmetry,^{55–57} on the coacervation behavior. With some fitting parameters, these models can produce qualitative or semiquantitative agreements with experimental data on the coacervate branch in the phase diagram. However, the dilute, supernatant phase has been relatively unexplored. Indeed, most theories for PE coacervation predict exceedingly low concentrations for the coexisting supernatant phase except near the critical point^{16,19,23–27,29–35} due to the assumption that the dilute phase is a uniform mixture of dispersed PE chains. However, the polycations and polyanions with the same chain structure

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and equal charge mostly exist as polyion pairs in the supernatant phase.^{28,58} The coacervate phase then forms from the aggregation of these polyion pairs and clusters via electrostatic correlation^{59,60} or dipole interaction.^{61,62} For a complete understanding of PE complex coacervation, it is necessary to better understand these basic structural units in dilute solution.

For polyion pairs formed from symmetric polycations and polyanions, the complex takes the form of a condensed spherical droplet, in which the polycation and polyanion are intertwined.⁶³ The polycation–polyanion complexes formed in asymmetric systems due to differences in chain length, concentration, etc., can take on more interesting structures. For example, Zhang and Shklovskii, using an energy argument, predicted that a long polyanion and many small polycations can form a “tadpole”-like complex.⁶⁴ Computer simulations later confirmed that the extreme chain-length asymmetry results in the formation of a loose, extended complex structure with net macromolecular charges.^{65,66} In multichain systems with symmetric polycations and polyanions of different concentrations, Monte Carlo simulations by Hayashi et al.^{67–69} on systems with relatively short chains (up to 20 monomers) with no salt or implicit salt showed that when the concentration ratio of the oppositely charged polyions is less than or equal to 0.5, the clusters are primarily of the “1 by 2” type (1 polyanion and 2 polycations in their case) with net macromolecular charge of the major component, while larger clusters were obtained at closer proportions between the polycation and polyanion. However, even the largest clusters in refs 67 and 69 remained quite loose and amorphous, and the system did not appear to form a coacervate phase even at the symmetric composition. Oskolkov and Potemkin,⁷⁰ using a combination of mean-field theory and random phase approximation, predicted that the polycation–polyanion complex with high concentration asymmetry can form cylindrical and lamellar structures. These pioneering studies on asymmetric polyelectrolyte mixtures reveal the richness in polyion clustering behavior compared to the simpler symmetric mixtures and motivate a more systematic investigation of the PE complexation in asymmetric mixtures in dilute solution and its connection to coacervation.

In this work, we examine the PE complexation behavior in mixtures with length or concentration asymmetry from the perspective of the dilute solution side by performing electrostatic dissipative particle dynamics (EDPD) simulations with explicit salt ions and solvent. For chain-length asymmetry, we start with a one-pair complex and systematically examine the evolution of the cluster structure as the chain-length ratio changes. We then investigate the clustering behavior in multichain systems with given chain-length asymmetries while maintaining the same number of polycations and polyanions. For concentration asymmetry, we set the chain length to be equal between the polycations and polyanions and change the proportion between the two kinds of polyions by adding the minor component to the major component until they reach equal concentration. We follow the clustering behavior for this titration process. Finally, we study the effects of added salts where we highlight the salt-induced aggregation and dissociation of the clusters under certain chain-length and concentration asymmetry.

SIMULATION METHOD

All simulations were performed with electrostatic dissipative particle dynamics (EDPD). DPD is a mesoscale coarse-grained method that is widely used in simulating polymers in solution^{71–76} because of its computational efficiency and ability to capture hydrodynamics due to the explicit account of solvent. Coupled with electrostatic interaction, EDPD is a powerful tool to simulate polyelectrolyte solutions,^{76–79} including PE complex coacervation systems.^{9,66} In our simulations, we set all the DPD beads to have the same mass m and a cutoff interaction distance r_c , which serve as the natural unit of mass and length, respectively.⁷¹ A DPD bead interacts with other beads through three types of short-ranged pairwise forces and the electrostatic force: $\mathbf{f}_i = \sum_j (\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R) + \mathbf{f}_i^E$. The sum is over all beads j within the cutoff r_c . The conservative repulsive force is given by $\mathbf{F}_{ij}^C = a_{ij}(1 - r_{ij}/r_c)\mathbf{e}_{ij}$, where a_{ij} is the strength of the interaction, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the interbead distance, and $\mathbf{e}_{ij} = (\mathbf{r}_i - \mathbf{r}_j)/r_{ij}$ represents the direction of the force. The dissipative force is related to the relative velocity between the two beads $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ via $\mathbf{F}_{ij}^D = -\alpha w_D(r_{ij})(\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij}$. The random force is of the form $\mathbf{F}_{ij}^R = \gamma w_R(r_{ij})\xi_{ij}\mathbf{e}_{ij}$, where $\gamma^2 = 2k_B T \alpha$ and ξ_{ij} a Gaussian random variable satisfying $\langle \xi_{ij}(t) \rangle = 0$ and $\langle \xi_{ij}(t)\xi_{kl}(t') \rangle = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(t - t')$. By the fluctuation–dissipation theorem, $[w_R(r_{ij})]^2 = w_D(r_{ij}) = (1 - r_{ij}/r_c)^2$.⁷¹ The thermal energy $k_B T$ is set to unity, which yields the characteristic time scale $\tau = \sqrt{mr_c^2/k_B T} = 1$. Nearest-neighbor beads along the polymer chain are subject to a harmonic bond potential $E_{\text{bond}} = 0.5K_{\text{bond}}(r - r_0)^2$ with $r_0 = 0.5r_c$ and a bending potential $E_{\text{angle}} = 0.5K_{\text{angle}}(1 + \cos \theta)$, where θ is the angle between two consecutive bonds. The bonding and bending interaction strengths are set to $K_{\text{bond}} = 128k_B T/r_c^2$ and $K_{\text{angle}} = 4k_B T$, respectively. We set $a_{ij} = 25k_B T/r_c$ for all DPD beads to represent good solvent condition for neutral polymers.

Long-range electrostatic force $\mathbf{f}_i^E$ in DPD is incorporated by using the particle-mesh method, as first developed by Groot.⁷⁶ Briefly, electrostatic interactions are calculated among the off-lattice charged beads by overlaying a square mesh with the simulation domain and the electric field is computed on the mesh. The point charge of a DPD bead is distributed to their nearby mesh nodes so that the center of partial charges on the nodes coincides with the bead position. The use of spread charge prevents the infinitely strong attraction between two opposite point charges because DPD only has soft short-range repulsion. Nevertheless, electrostatic DPD has been shown to be able to capture strong correlation effects in polyelectrolyte solution systems, such as counterion condensation.⁷⁷ The Poisson equation on the mesh is solved by the finite difference discretization and the real-space successive over the relaxation (SOR) method. The red–black SOR method is applied to parallelize the Poisson solver for simulating large-scale systems. Details of the implementation have been described elsewhere.^{76,80}

The polyelectrolytes in our simulations are linear chains with every monomer carrying unit charge $+e$ or $-e$. Each charged polymer bead has the corresponding counterion bead carrying the opposite unit charge. Salt ions are modeled as monovalent beads having the same identity as the counterions. The Bjerrum length given by $l_b = e^2/4\pi\epsilon_0\epsilon_w$, where ϵ_0 and ϵ_w are the vacuum permittivity and the relative dielectric constant of water at room temperature, respectively, is taken to be 0.7 nm.

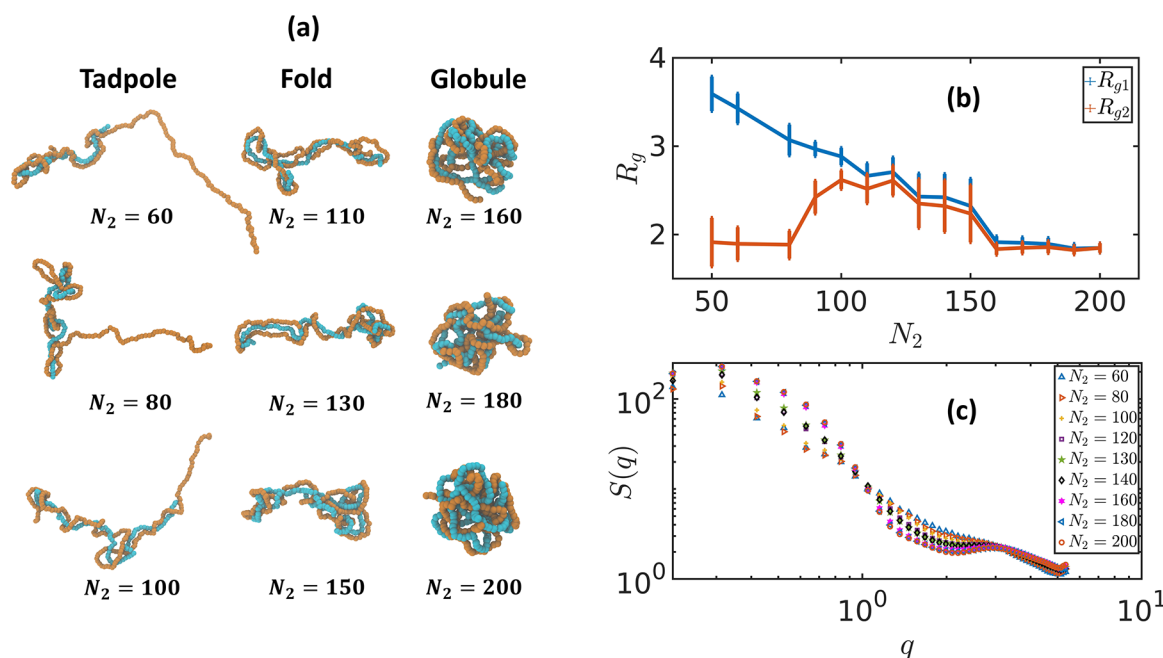


Figure 1. (a) Morphology of a polyion pair under different length asymmetry. The polyanion chain length is fixed at $N_1 = 200$, while polycation chain length varies with $N_2 = 50$ –200. (b) Radius of gyration of the two chains at different chain length ratios. The error bar represents standard deviation in 100 equilibrium simulation snapshots. (c) Structure factor of the long polyanion chain. The polyanion is colored in orange, and the polycation is colored in cyan. Water beads are not shown for clarity.

The electrostatic interaction strength can be characterized by $\Gamma = l_b/r_c$. With the typical interpretation that one DPD bead corresponding to three water molecules,⁷⁶ we have $\Gamma \approx 1.1$. For the single polyion pair study in the case of chain length asymmetry, we fix the length of polyanion at $N_1 = 200$ and vary the length of the polycation in range $N_2 = 50$ –200, with increments of 10. For the examination of the chain-length asymmetry in the multichain systems, we set the polyanion length to be $N_1 = 50$ and vary the polycation length in the range of $N_2 = 25$ –50. For the concentration asymmetry study, we fix $N_1 = N_2 = 50$ and vary the concentration ratio $x_c = 0.1$ –1.0. In the remainder of the paper, we use the subscript 1 (2) to refer to the longer (shorter) chain or the major (minor) component. Small counterions for each polyion are added to maintain the global charge neutrality of the system. The simulation box has dimensions of $60r_c \times 60r_c \times 60r_c$ with the periodic boundary condition in all three directions. The simulation time step is 0.04τ . All simulations are performed in the NVT ensemble. The simulation time ranges from 0.5×10^6 to 4×10^6 time steps (corresponding to physical time about 0.8×10^5 to 6.4×10^5 ns). To characterize the cluster distribution in the multichain systems, we shall adopt the nomenclature where the clusters are termed “ $a \times b$ ” clusters, with “ a ” being the number of chains of the minor component (the polycation) and “ b ” the number of chains of the major component (the polyanion) in a cluster.

RESULTS AND DISCUSSION

Chain-Length Asymmetry. We first consider a single pair of oppositely charged chains in a salt-free system. Figure 1a shows the representative equilibrium structures of the polyion pair under different chain-length asymmetry. We observe three types of morphology: When the chain length of the shorter polycation (represented by cyan in panel a) is less than approximately half that of the long polyanion (represented by

orange), i.e., $x_1 = N_2/N_1 < 0.5$, a “tadpole”-like cluster is observed in which the shorter polycation binds to one end of the long polyanion while leaving the rest of the polyanion chain bare. The structure is similar to the “tadpole” morphology predicted by Zhang and Shklovskii⁶⁴ for a system of a single long polyanion with many short polycations where the total charge of polycations is much less than that of the long polyanion. However, while the “tadpole” structure predicted in ref 64 represents a neutral condensed head, the “tadpole” head in our two-chain system is overcharged by the longer polyanion; that is, the shorter polycation binds with a polyanion section longer than the length of the polycation. This feature is especially obvious for the case of $N_2 = 100$. In spite of the difference in the structure and charge in the condensed region between our observed structure and that predicted in ref 64, we shall adopt the term “tadpole” proposed by these authors for easy reference. We note that the “tadpole” structure was also predicted in the self-coacervation of a block polyampholyte.^{29,81,82} The tadpoles formed in these block polyampholytes usually have a globular head, probably due to differences in solvent quality, block connectivity, charge density, and degrees of charge asymmetry. The tadpole structure obtained in our simulation is more similar to what Jeon and Dobrynin⁸³ observed in their Monte Carlo simulation of the complexation of a short polyampholyte with a long polyanion in that the “tadpole” head in their case also has an extended structure and is negatively overcharged. The overcharge in our case can be rationalized with the following consideration: by having extra chain lengths of the polyanion in the vicinity of the shorter polycation, the electrostatic interaction energy between the polycation and polyanion beads can be further decreased without incurring much penalty due to the repulsion between the polycation beads. This effect becomes most pronounced in the intermediate regime with $0.5 < x_1 < 0.8$, when the entire long polyanion wraps along the

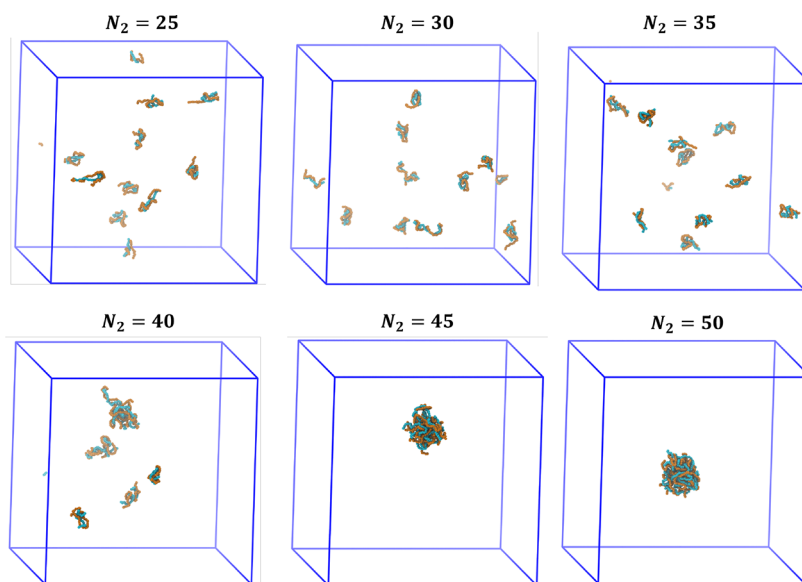


Figure 2. Evolution of the cluster structure with the chain length ratio for the case of chain-length asymmetry. The system contains 10 polycations and 10 polyanions. The polyanion length is fixed at $N_1 = 50$, while the polycation length takes $N_2 = 25, 30, 35, 40, 45$, and 50 . There are 10 clusters with 1×1 type with $N_2 = 25, 30$, and 35 , five clusters of different sizes (three 1×1 , one 2×2 , and one 5×5) with $N_2 = 40$, and one single cluster with $N_2 = 45$ and 50 . Water beads are not shown for clarity.

shorter polycation, forming an extended “fold”-like cluster, as shown in the middle column of Figure 1a. Interestingly, in this regime, the physical dimension of the shorter polycation, as measured by its radius of gyration, exhibits a nonmonotonic change with the length of the shorter chain. With further increasing the short chain length to ($x_1 > 0.8$), the two chains collapse to a globule-like cluster, consistent with previous studies.^{63,66} We note that the nonmonotonic behavior in the radius of gyration of the shorter chain in the polyion duplex was also reported in ref 66, although that work did not explicitly report the fold-like morphology.

As the chain length of the shorter polycation increases, the conformation of the long polyanion chain undergoes considerable changes over the three types of morphology. To further quantify the chain conformation change of the polyanion beyond the overall radius of gyration (as shown by the blue curve in Figure 1b), we compute its structure factor defined by $S(q) = \langle \hat{\rho}_q \hat{\rho}_{-q} \rangle / N$, where $\hat{\rho}_q$ is the polymer bead density in Fourier space given by $\hat{\rho}_q = \sum_i e^{-i\mathbf{q} \cdot \mathbf{r}_i}$. Here, $\mathbf{q}$ is the wavevector and $\mathbf{r}_i$ is the position vector of bead i ; i is the unit imaginary number. For very long chains in the absence of structural heterogeneity, the structure factor should scale as $S \sim q^{-1/\nu}$ in the intermediate range of q , corresponding to the chain size scaling $R_g \sim N^\nu$. Because the chain length is only moderately long in our simulation and because of the heterogeneity in the different parts of the polyanion in the tadpole regime, we do not observe a clear power-law scaling. Nevertheless, the steeper slope in the log–log plot with increasing length of the shorter polycation is consistent with the conformation of the complex becoming more compact.

It is natural to ask how the complexation behavior changes when there are multiple chains in the system. To this end, we consider a mixture of $n_1 = 10$ polyanions and $n_2 = 10$ polycations, with a fixed polyanion chain length $N_1 = 50$, and varying the polycation length $N_2 = 25, 30, 35, 40, 45$, and 50 . The equilibrium structures are shown in Figure 2. In systems with $N_2 \leq 35$, we find that in the majority of cases each

polycation pairs with one polyanion, forming individual 1×1 pairs that carry net negative charges. These 1×1 complexes remain dominant in a significant range of chain length asymmetry, indicating that the system will be in the state of a solution of these charged pairs as the basic structural units. At $N_2 = 40$, larger clusters of several different sizes begin to form. The number of clusters change dynamically, such that small pairs can merge into and separate from the larger clusters. The most popular cluster distribution in this system is the coexistence of three 1×1 clusters, a 2×2 cluster, and a 5×5 cluster. When N_2 increases to 45 corresponding to $x_1^* \approx 45/50 = 0.9$, all the PE chains condense into a single large (9×10) cluster. These behaviors are qualitatively unchanged when we increase the number of polyions to $n_1 = n_2 = n = 50$ (see Figure 6b): when $x_1 \lesssim 0.7$ (for the chain lengths ratio studied in our system), the system contains predominantly 1×1 pairs, while for $x_1 \gtrsim 0.9$ all the polyions condense into a single large cluster. The main difference for these larger systems is that there are more cluster varieties in the range $0.7 \lesssim x_1 < 0.9$. The emergence of a single large cluster in the simulation for the multichain mixtures is indicative of the existence of a coacervate phase. Thus, the threshold value x_1^* at which many small clusters merge into a large cluster should mark the chain-length asymmetry condition for the macroscopic phase transition. However, the system size here is too small to determine x_1^* , as in a macroscopic system we expect a dominant large cluster coexisting with a dilute phase of small PE clusters; in later sections we will examine larger system sizes. For the chain lengths we have examined, all the polyions are almost always in some kinds of clusters, although naked (uncomplexed) chains have occasionally been observed for short periods of time. However, we should note that the polycation and polyanion lengths are still comparable even when they are not equal. Different behavior can be expected with extreme chain length asymmetry; for example, very short polycations can behave more like small counterions. Such extreme cases are beyond the scope of our current work.

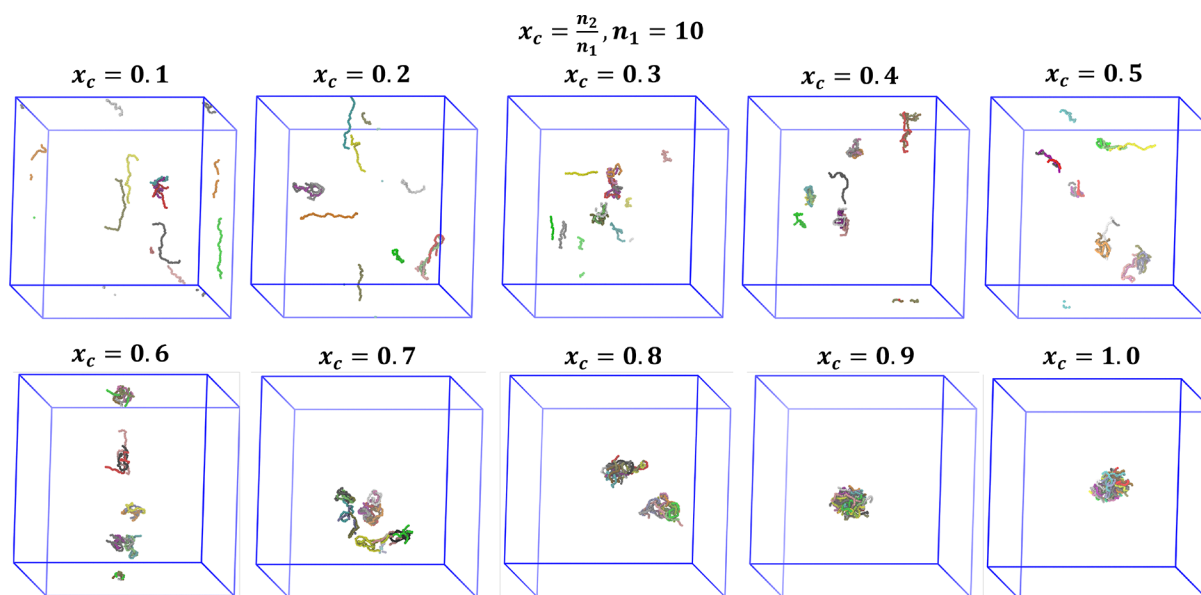


Figure 3. Polyion clustering behavior in dilute solution with charge ratio $x_c = n_2/n_1$. The number of polyanions is fixed at $n_1 = 10$. When $x_c \lesssim 0.5$, the only clusters are of the type 1×2 , along with excess polyanions. For $x_c \gtrsim 0.5$, there are no uncomplexed polyanions. In the range between 0.6 and 0.8, clusters exist in the form of 1×2 , 2×3 , 3×4 , and 4×5 .

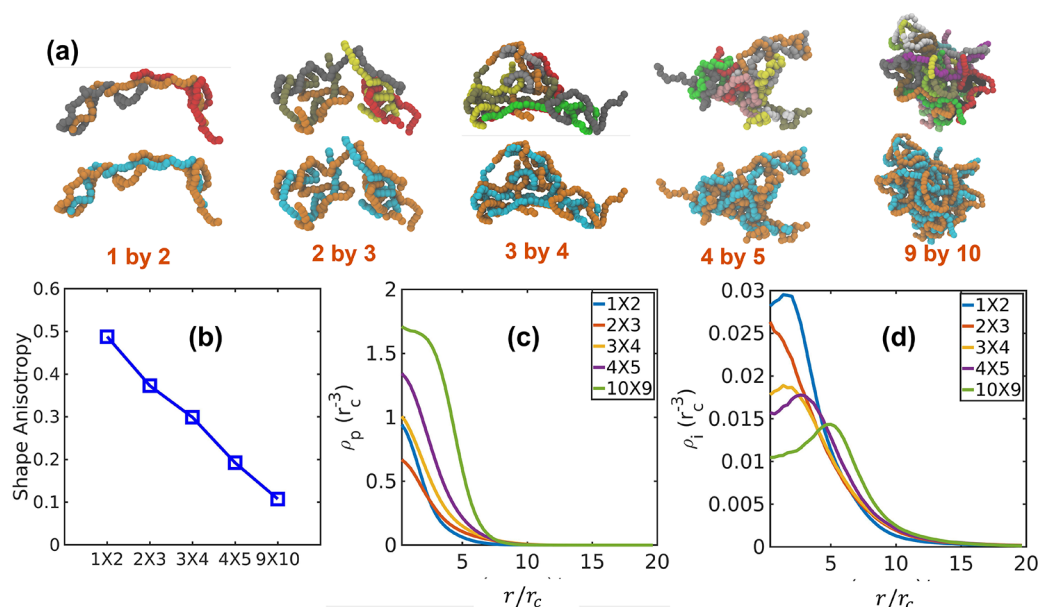


Figure 4. (a) Snapshots of clusters for concentration asymmetry. In the first row, the different chains have different colors. For comparison, in the second row, the polycations are colored in cyan and polyanions are colored in orange. (b) Shape anisotropy for the different cluster types. (c) Radial monomer density profile for each net-charged cluster. (d) Net counterion charge density profile for each cluster.

Concentration Asymmetry. In this section, we consider the effects of concentration asymmetry on PE complexation under salt-free conditions. We first consider a mixture of $n_1 = 10$ polyanions and $n_2 = 1$ –10 polycations, with the same chain length of all polyions fixed at $N_1 = N_2 = N = 50$. For strongly asymmetric systems with charge ratio $x_c = n_2/n_1 \lesssim 0.5$, the dominant cluster structure is of the 1×2 type, i.e., one polycation complexing with two polyanions, as shown in the first row of Figure 3. The excess polyanions remain uncomplexed. As we increase x_c to the range 0.6–0.9, larger net-charged clusters with varying stoichiometries are formed. The clusters can take the form of 1×2 , 2×3 , 3×4 , 4×5 , and 9×10 . In this range of asymmetry, all polyions are in

some kinds of complexes, and all complexes are charged with excess polyanions. When $x_c \gtrsim 0.9$, all polyions aggregate into one single globular cluster.

The predominance of the 1×2 cluster in strongly asymmetric systems with equal polycation and polyanion lengths indicates that the most asymmetric cluster in this case is one polycation complexing with two polyanions. The overcharge of this cluster is a consequence of the excess of polyanions in the system. A polyanion can lower its electrostatic energy by complexing with a polycation–polyanion pair. However, further overcharging would induce too much net charge in the cluster. This strong preference for the 1×2 clusters in systems with high concentration

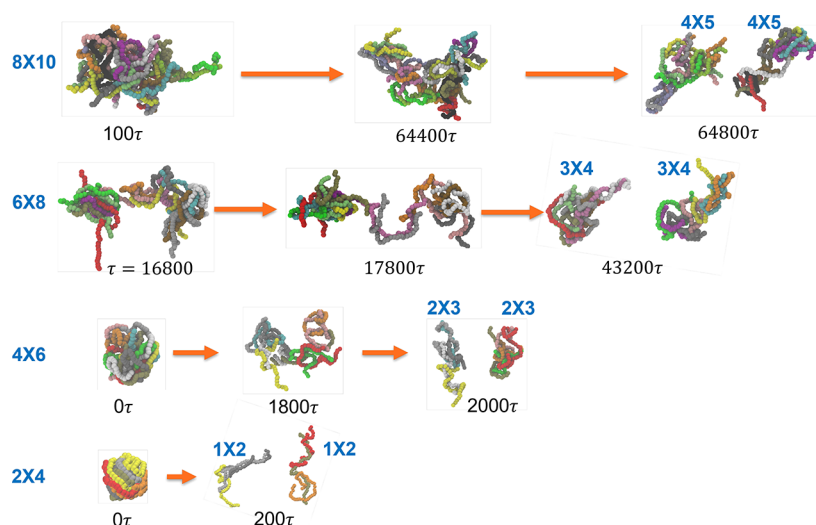


Figure 5. Snapshots for the fission of unstable $2a \times 2b$ clusters. τ is the elementary DPD time unit.

asymmetry is in agreement with earlier results from Monte Carlo simulations of systems of shorter chains ($N_1 = N_2 = 20$).⁶⁷ We have further tested the robustness of this result with chain lengths up to $N_1 = N_2 = 140$ (see Figure S1).

As in the case of chain-length asymmetry, we believe the emergence of a single large cluster at $x_c \gtrsim 0.9$ suggests the existence of a threshold value x_c^* which marks the onset of the macroscopic coacervate transition; this is consistent with the theoretical prediction that coacervation in salt-free systems can only take place when the polycation–polyanion concentration asymmetry is not high.⁵⁵ We will discuss the threshold value x_c^* using larger system size where we can see a dominant large cluster coexisting with smaller clusters. We note that the transition from multiple clusters to a single large cluster at $x_c = 0.9$ – 1.0 was not observed in previous Monte Carlo studies.^{68,69} Instead, the tendency to form large clusters was seen to reach a maximum at $x_c = 0.9$ in these works, with a large cluster coexisting with some smaller clusters; further increasing the ratio to equal numbers of polycations and polyanions resulted in multiple small neutral clusters. This discrepancy is likely due to the use of screened, cutoff Coulomb interactions in these works, which underestimated the cohesive force due to electrostatic correlation, as pointed out by these authors.⁶⁷ Interestingly, for all systems with concentration asymmetry up to $x_c = 0.9$ examined in our study, we do not find any symmetric (e.g., 1×1 , 2×2 , etc.) clusters. This is in contrast to the case of chain-length asymmetry, where we obtain 1×1 pairs in a broad range of length ratios. The different complexation behaviors between chain-length and concentration asymmetry indicate there is a fundamental difference between these two kinds of asymmetry in the dilute phase.

Properties of the Net-Charged Clusters. In this section, we take a closer look at several properties of the net-charged clusters, focusing on the case of concentration asymmetry.

We first consider the shape anisotropy of these clusters. To this end, we perform five simulations where each system contains only one net-charged cluster (with its counterions) for a given stoichiometry (1×2 , 2×3 , 3×4 , 4×5 , 9×10). We compute the shape anisotropy defined as $\langle 3(R_g^4 + R_g^2 + R_g^4)/[2(R_g^2 + R_g^2 + R_g^2)] \rangle - 1/2$, where R_g^2 is the a th eigenvalue of the square radius of gyration tensor. A perfect sphere has zero anisotropy while a rod with a large aspect ratio

has nearly unit anisotropy. Figure 4a shows some representative simulation snapshots for the five studied clusters (1×2 , 2×3 , 3×4 , 4×5 , and 9×10). The corresponding anisotropy is shown in Figure 4b. Clearly, as the stoichiometry in the cluster becomes more symmetric, the clusters become more spherical. As a point of comparison, we note that an ideal Gaussian chain has a shape anisotropy of $10/19 \approx 0.53$,⁸⁵ which is larger than the most nonspherical cluster (1×2) whose shape anisotropy is around 0.49. In general, the clusters are more compact as the stoichiometry becomes more symmetric. This is reflected in the monomer density distribution shown in Figure 4c as a function of the radial distance r from the center of mass of the cluster. Overall, the monomer density is higher for the more symmetric clusters. A special case is the 2×3 cluster, whose structure shows a dumbbell-like shape, with two 1×1 pairs at each side bridged by an extra polyanion. This morphology makes the monomer density lower at the center and span a larger distance than the more asymmetric 1×2 cluster.

Because these clusters carry net macromolecular charges, they are expected to be surrounded preferentially by the small counterions of opposite charge. In Figure 4d we show the net-charge density profile of the small counterions, defined as $\rho_i = \rho_{\text{cation}} - \rho_{\text{anion}}$, where ρ_{cation} and ρ_{anion} are the density of the small cations and small anions, respectively. As expected, ρ_i is higher around more asymmetric clusters, as they carry higher macromolecular net charges. Interestingly, the counterion distributions show a peak slightly outside of the PE cluster; this feature is in agreement with a recent theoretical prediction.⁸⁶ The much broader counterion charge distribution compared to that of the polyion cluster further confirms the overall net-charged nature of these clusters.

The clusters shown in Figure 3 are obtained by titrating a polyanion solution with increasing numbers of polycations. To ensure that these structures are thermodynamically stable and are not a result of this particular preparation pathway, we perform a set of simulations by starting with larger cluster sizes of the form $2a \times 2b$, where a and b are the number of polycations and polyanions in the clusters shown in the first four structures in Figure 4a. If a $a \times b$ net-charged cluster is thermodynamically stable in dilute solution, a $2a \times 2b$ cluster will be unstable and should split into two ($a \times b$) clusters or

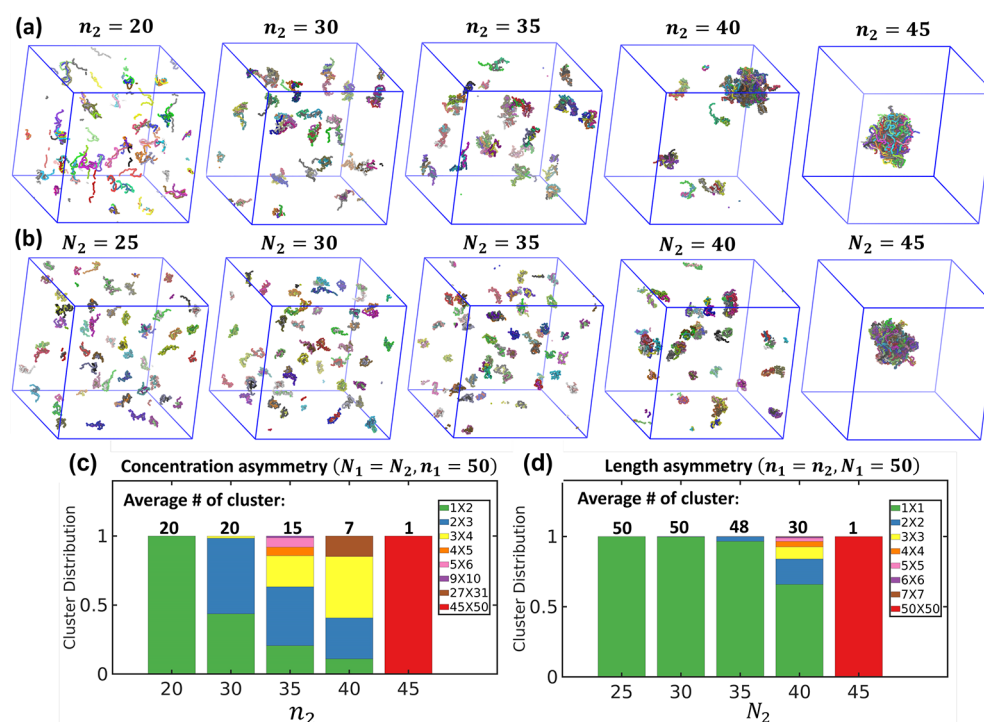


Figure 6. (a) Snapshots of system configuration for concentration asymmetry with charge ratio $x_c = n_2/n_1 = 0.4, 0.6, 0.7, 0.8$, and 0.9 ; the number of polyanions is fixed at $n_1 = 50$. (b) System configuration for chain-length asymmetry with charge ratio $x_l = N_2/N_1 = 0.5, 0.6, 0.7, 0.8$, and 0.9 . The chain length of the polyanion is fixed at $N_1 = 50$. (c) Cluster distribution in (a). (d) Cluster distribution in (b).

other smaller-cluster combinations. We prepare these double-sized clusters by mixing a given number of chains with their counterions under poor solvent conditions without including electrostatic interactions. After the chains form a well-mixed globule, we turn on the electrostatic interactions and start the clock. In Figure 5, we show the fission of the double-sized clusters and some intermediate states. The 2×4 is highly unstable and immediately splits into two 1×2 (the time shown is in the DPD time units). As the asymmetry decreases, the fission time increases. The fission is driven by the decrease in the electrostatic energy in going from a single overcharged cluster to two smaller clusters each carrying less charge. Figure S2 shows the evolution of the total electrostatic energy along the fission pathway of the 2×4 cluster to the two 1×2 clusters. The large energy drop (nearly $100 k_B T$) explains the rapidity of the process. For the less asymmetric clusters, the energy decrease is less (not shown) and the fission process takes longer. The symmetric splits shown in Figure 5 are the dominant outcomes. The larger, double-sized clusters can occasionally split unevenly. For example, the 6×8 can also split into a 4×5 and a 2×3 (not shown).

Cluster Distributions. Our results so far have revealed the existence and properties of the net-charged PE clusters in a very dilute system with total PE chains less than 20. To obtain meaningful information about the distribution (i.e., their frequency of occurrence) and to have a better estimate of the threshold charge ratio for coacervation, a larger system size with more cluster varieties is needed. To this end, for concentration asymmetry, we perform simulations at a larger number of polyanions $n_1 = 50$, while varying the number of polycations $n_2 = 20, 30, 35, 40, 45$, and 50 , corresponding to concentration ratios $x_c = n_2/n_1 = 0.4, 0.6, 0.7, 0.8, 0.9$, and 1.0 . For chain-length asymmetry, we consider systems with equal number of polycations and polyanions $n_1 = n_2 = 50$ with fixed

polyanion length at $N_1 = 50$ and a series of polycation lengths $N_2 = 25, 30, 35, 40$, and 45 , corresponding to length ratios $x_l = 0.5, 0.6, 0.7, 0.8$, and 0.9 . We now discuss the results for these two different asymmetries separately.

Figure 6a shows snapshots of the systems with concentration asymmetry. A clear trend is that the number of clusters decreases as the compositions become more symmetric. In Figure 6c, we show the cluster distribution of different types as a function of the concentration ratio x_c . The cluster distribution for both types of asymmetry is calculated by averaging over 1000 snapshots in the latter half of the simulation run with 10^3 time steps between two neighboring snapshots. For high concentration asymmetry $x_c = 0.4$, the only clusters are of the 1×2 type. At x_c increases, slightly larger clusters types of 2×3 and 3×4 start to emerge and can overtake the population of the 1×2 clusters. The richest cluster variety takes place at $x_c = 0.7$; further increasing the concentration of polycations to $x_c = 0.8$ results in the formation of a much larger cluster (27×31) that coexists with some small clusters. We believe the emergence of this dominant cluster in our systems with finite box size signals the onset of macroscopic coacervation in the bulk systems. This means the threshold value for concentration asymmetry in salt-free conditions is around $x_c^* \approx 0.8$. Finally, all polyanions condense into a single droplet at $x_c \gtrsim 0.9$.

Figures 6b and 6d show respectively snapshots of the clusters and their distribution for the case of chain-length asymmetry. Interestingly, the 1×1 type cluster dominates the cluster population for all length ratios prior to $x_l = 0.9$ when all the polyanions condense into a single cluster. Furthermore, all the clusters are of the $a \times a$ type, i.e., clusters have equal numbers of polycations and polyanions, even though the clusters carry a net charge. The richest cluster varieties happen at $x_l^* = 0.8$. In contrast to concentration asymmetry systems, at

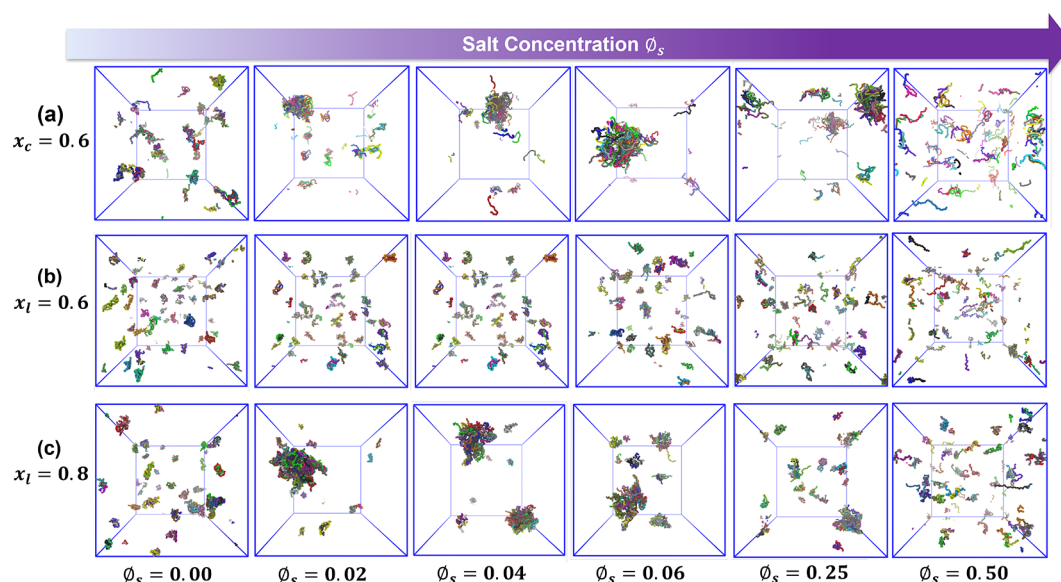


Figure 7. Effects of added salt $\phi_s = 0.00, 0.02, 0.06, 0.25$, and 0.50 on the clustering behavior in concentration and chain-length asymmetry systems: (a) concentration asymmetry at $x_c = 0.6$, (b) chain length asymmetry at $x_l = 0.6$, and (c) chain length asymmetry at $x_l = 0.8$. In all cases, the major component has $n_1 = 50$ chains and is of length $N_1 = 50$.

the same total charge ratio, the chain-length asymmetry systems have a tendency to contain uniform, small clusters, with the 1×1 type dominating the cluster population. Also, in the case of concentration asymmetry, the clusters are always of the type $a \times b$ with $a \neq b$. Interestingly, at charge ratio $x_l = 0.8$, there is no clear sign of the emergence of a dominant large cluster, in contrast to the case of concentration asymmetry at the same charge ratio $x_c = 0.8$. This difference indicates that in salt-free systems the threshold value that signals the onset of macroscopic coacervation is less for concentration asymmetry than for the chain-length asymmetry system. From Figure 6, we find $0.8 \approx x_c^* < x_l^* \lesssim 0.9$.

To understand the striking difference in the cluster distribution between the two kinds of asymmetry, we note that in the case of chain-length asymmetry with equal number of polycations and polyanions the system starts with 1×1 clusters at small x_l , as mentioned in our discussion of Figure 2. These uniform, equal-charge small clusters evenly distribute the net macromolecular charge of the systems; an uneven distribution would leave some clusters with too much net charge. Such an energetically favorable distribution is maintained until a rather high x_l when clusters of the form $a \times a$ ($a > 1$) begin to emerge as a result of the merging of the 1×1 clusters. However, the energetic driving force for merging 1×1 clusters into larger ones is not very strong as it consists of two competing effects—the tendency to decrease the surface energy (which favors merging) and the increased overcharge in the larger cluster (which disfavors merging). This relatively weak driving force together with the larger translational entropy of the 1×1 clusters makes them the dominant population until $x_c \approx 0.9$. On the other hand, for concentration asymmetry, the system starts with 1×2 clusters at x_c (< 0.5) to minimize the number of uncomplexed polyanions without incurring too much net (negative) charge in the cluster. As $x_c = 0.5$, all the polyions are in the form of 1×2 clusters. Further increasing x_c will lead to the formation of less asymmetric clusters, such as 2×3 , 3×4 , etc., and the cluster distribution keeps shifting toward larger clusters with more symmetric compositions.

Overall, for both chain-length and concentration asymmetry, larger clusters are formed when decreasing the overall charge asymmetry between polyanions and polycations. However, at the same level of charge asymmetry, systems with concentration asymmetry tend to form $a \times b$ clusters with $a < b$ so as to minimize the number of naked polyions, while systems with chain-length asymmetry usually form small $a \times a$ clusters so as to evenly distribute the charges. Consequently, at the same charge ratio, systems with concentration asymmetry have the tendency to form larger clusters and are more prone to phase separation, while systems with chain-length asymmetry will have a larger window of stability as a homogeneous phase.

Effects of Added Salt. Our study of the salt-free systems examined so far reveals a wealth of interesting structural properties, particularly with respect to the shape, size, and composition of the polyion clusters. The formation and distribution of these net-charged clusters are of course a result of the electrostatic interactions and correlations, which can be modulated by the addition of salt. It is therefore of interest to study how addition of salt ions affects the clustering behavior. To this end, we first pick two salt-free systems from Figures 6a and 6b with the same total charge asymmetry $x_c = x_l = 0.6$ and the same total monomer density for concentration asymmetry ($N_1 = N_2 = 50$, $n_1 = 50$, $n_2 = 30$) and chain-length asymmetry ($n_1 = n_2 = 50$, $N_1 = 50$, $N_2 = 30$). We then add salts with total volume fraction (cations and anions) in the range $\phi_s = 0.02$ – 0.50 (corresponding to salt concentration $c_s = 0.17$ – 4.50 M). (The behavior at $\phi_s = 0.01$ was found to be similar to the salt-free case.)

Figure 7a shows the clustering behavior for the case of concentration asymmetry at $x_c = 0.6$. The general trend is that as the salt concentration increases, the average cluster sizes initially increases while their number decreases. Because the total macromolecular charge ratio is fixed, the decrease in the number of small clusters of the 1×2 type is accompanied by the emergence of larger clusters that can be quite asymmetric (e.g., 23×33 in the second figure of panel a at $\phi_s = 0.02$ in Figure 7). The size of the dominant cluster grows at the expense of the smaller clusters, as the added salt increases to ϕ_s .

= 0.04 and $\phi_s = 0.06$. As discussed before, we believe the emergence of a dominant cluster in our simulation with finite box size is indicative of the onset of macroscopic coacervation in the bulk system. As the salt concentration varies in increment of 0.01, we are unable to precisely pinpoint the value of the salt concentration at which this condensation happens, but it is around $\phi_s = 0.02$ from visual inspection. Compared to the salt-free systems where the threshold value is around $x_c^* \approx 0.8$, coacervation can take place at a lower charge ratio ($x_c = 0.6$ here) with added salt. As the salt concentration is further increased to $\phi_s = 0.25$, the large cluster begins to dissolve, and at even higher salt concentrations, the larger cluster completely disintegrates into many small clusters and even some uncomplexed polyanions. The evolution of the weight-averaged cluster size (normalized by the total weight of the polyions) is shown in Figure S3. The merging of small clusters with increasing salt at lower salt concentration can be understood by the screening of the repulsion between the net-charged small clusters and the increased electrostatic correlation which favors coalescence, while too much salt decreases the driving force for complexation between the oppositely charged polyions, resulting in a more dispersed small cluster and even bare polyanions. This behavior of “salting out” followed by “salting in” with increasing salt concentration is consistent with recent findings in the combined experimental and theoretical work in ref 56 and the more recent computer simulation study in ref 57. The results presented here highlight the connection between clustering in dilute solution and macroscopic coacervation.

Interestingly, for chain-length asymmetry at $x_1 = 0.6$, which is the same macromolecular charge ratio as for the case of concentration asymmetry shown in Figure 7a, we do not observe obvious “salting out” behavior with increasing salt. Instead, Figure 7b shows that the system prefers to have dispersed (mostly 1×1) clusters for all salt concentration conditions until many of the polyions become uncomplexed at $\phi_s \approx 0.50$. The “salting out” to “salting in” transition, however, can take place when the ratio of polycation length to polyanion length is increased to $x_1 = 0.8$ ($N_2 = 40$), as shown in Figure 7c. This coacervation behavior at charge ratio $x_1 = 0.8$ with added salt at $\phi_s = 0.02$, which is not seen in the corresponding salt-free system, again confirms that the added salt can bring down the threshold charge ratio x_1^* . The weight-averaged size of the clusters for all the three systems shown in Figure 7 is given in Figure S3. The difference in behavior between the concentration asymmetry and length asymmetry for the same macromolecular charge ratio $x_c = x_1 = 0.6$, as shown in Figures 7a,b and Figure S4, indicates that for a given total charge asymmetry, the small(er) net-charged clusters are more stable against added salt when the asymmetry is in the chain length than in concentration. As discussed for the salt-free systems (see Figure 6c,d), the 1×1 clusters in the chain-length asymmetry systems are favored both energetically (because of the even distribution of the net charges) and entropically (because of their smaller size). Thus, “salting out” in chain-length asymmetry systems requires a larger x_1 compared to the concentration asymmetry systems.

CONCLUSION

In this work, we have performed EDPD simulations to systematically investigate the polyelectrolyte complexation behavior in dilute solution for systems with length and concentration asymmetry between the polycation and the

polyanion. Our results show that generally charge asymmetry leads to the formation of net-charged polycation–polyanion clusters carrying the sign of the major component. We have examined the morphology, composition, stability, and size distribution of these clusters and the salt effects on these properties under various chain-length and concentration asymmetry conditions. In salt-free systems, the complexation of one polycation and one polyanion can take the morphology of “tadpole”, “fold”, and “globule” under different chain-length asymmetry ratios. In multichain systems, stoichiometry and distribution of the clusters are very different between the two types of asymmetry: For the case of concentration asymmetry, the 1×2 type complex dominates the cluster population for $x_c \lesssim 0.5$. Further increasing x_c results in the emergence of larger and less asymmetric clusters, e.g., 2×3 , 3×4 , 4×5 , etc. When x_c reaches a threshold value $x_c^* \approx 0.8$, most of the polyions condense into a single dominant cluster, signaling the onset of macroscopic coacervation. For the case of chain-length asymmetry, small 1×1 polyion pairs dominate the cluster population for all length ratios prior to $x_1 \approx 0.9$ when all polyions condense into a single cluster. Moreover, systems with chain-length asymmetry with equal number polycations and polyanions always form clusters of the $a \times a$ type. The threshold value in x_c and x_1 decreases with added salt, with the case of chain-length asymmetry requiring a larger x_1^* than x_c^* . For sufficiently asymmetric mixtures in the case of length asymmetry, condensation is not observed for salt concentrations up to $\phi_s = 0.5$ (corresponding to 4.50 M). For systems where condensation can take place ($x_c = 0.6$ for concentration asymmetry and $x_1 = 0.8$ for chain-length asymmetry), further adding salt leads to the dissolution of the large cluster into smaller clusters and even uncomplexed polyions. This “salting-out” followed by “salting in” behavior is consistent with the recent experimental observation in ref 56 and computer simulation results in ref 57.

While the focus of our work is on the clustering behavior of polycations and polyanions in dilute solution, our results have interesting implications for the macroscopic coacervation behavior. For example, for the concentration-asymmetry case at $x_c = 0.6$, upon the formation of a large cluster with the addition of salt (at $\phi_s \approx 0.02$), the overall concentration of polyions outside the large cluster decreases upon further increasing salt concentration, implying that the dilute branch in the polyion–salt phase diagram has a negative slope. This feature is actually consistent with the slope of the experimentally measured phase boundary on the dilute side reported in ref 56 at low salt concentrations. Also, the prevalence of net-charged clusters at high charge asymmetry suggests that at low salt concentrations a polyelectrolyte mixture may form a structured fluid composed of finite-sized clusters of some preferred length scale. In addition, the clustering behavior is likely to be impacted by the solvent quality. In this work, we have considered the case of good solvent. Decreasing the solvent quality should increase the propensity for clustering, thus affecting the cluster size distribution, possibly the composition, and ultimately the threshold for macroscopic condensation. We plan to examine these issues in future studies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.2c00339>.

Polyion clustering in dilute solution with one polycation and five polyanions, for increased chain lengths $N_1 = N_2 = 80, 100$, and 140 ; evolution of the electrostatic energy (including self-energy) during the fission of a 2×4 clusters into two 1×2 clusters; dependence of the weight-averaged mass M_w of clusters in systems of $x_c = 0.6$, $x_1 = 0.6$, and $x_1 = 0.8$ on the concentration of added salt (PDF)

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Notes

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